Calculus II (MATH 162)

Midterm Exam (Fall, 2025)

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풀이 과정을 자세히 기술해야 합니다.

1. (10pts) A lamina occupies the region inside the circle $x^2+y^2=2y$ but outside the circle $x^2+y^2=1$. Find the center of mass of the lamina that occupies the region if the density function is

$$\rho(x,y)=\sqrt{x^2+y^2}$$

2. (10pts)

(a) Find the local maximum and minimum values and saddle point(s) of the function

$$f(x,y)=x^3+3xy^2-15x+y^3-15y$$

(b) Convert the integral to an equivalent integral in polar coordinates.

$$\int_0^1\int_{\sqrt{1-y^2}}^{2} f(x,y)\,dx\,dy \quad +\int_1^{2\sqrt{3}}\int_0^{2} f(x,y)\,dx\,dy$$
$$+\int_{2\sqrt{3}}^{4}\int_0^{\sqrt{16-y^2}} f(x,y)\,dx\,dy$$

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3. (10pts) Find the limit, if it exists, or show that the limit does not exist.

(a) $\lim_{(x,y)\to(0,0)} \dfrac{x^3\sin^2 y}{(x^2+y^2)^2}$

(b) $\lim_{(x,y)\to(1,0)} \dfrac{\sin(xy^2)\cos(y^2)-\cos(xy^2)\sin(y^2)}{(x-1)^2+2y^4}$

4. (10pts) Let E be the region bounded by $z=(x^2+y^2)^2$, $z=x^2-y^2$ and xz-plane in the first octant. Evaluate the integral

$$\iiint_E 2xy\,dV$$

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5. (10pts) Assume that $z=f(x,y)$ has continuous second-order partial derivatives. If $x=\frac{t}{s}$, $y=\frac{s}{t}$, $f_x(1,1)=2$, and $f_y(1,1)=1$, then find

$$s^2\frac{\partial^2 z}{\partial s^2}-t^2\frac{\partial^2 z}{\partial t^2}$$

when $s=2$ and $t=2$.

6. (10pts) Let $F(x,y,z)$ be a differentiable function where $\frac{\partial F}{\partial z}\neq 0$, and let $z=z(x,y)$ be a differentiable function defined by $F(x,y,z(x,y))=0$.
Suppose that $2x+5y-2z=5$ is the tangent plane to the surface $F(x,y,z)=0$ at the point $(1,1)$.
(a) Find the rate of change of $z(x,y)$ in the direction $\mathbf{v}=\langle 2,3\rangle$ at $(1,1)$.
(b) Use linear approximation to estimate $z(1.1,0.9)$.

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7. (10pts) Let $f(x,y)=e^{\left(x-\frac{1}{2}\right)^2+y^2}$. Find the absolute maximum and minimum values of $f(x,y)$ on the region

$$D=\left\{(x,y): x^2+y^2 \le \sqrt{x^2+y^2}+x\right\}$$

8. (10pts) Use Lagrange multiplier method to find the extreme values of the function $f(x,y,z)=x^2+y^2+z^2$ on the curve of intersection of the surface $x^2+y^2+z^4=1$ and the plane $y=x$.

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9. (10pts) Evaluate the integral

$$\int_0^2 \int_0^{1-\frac{y}{2}} \int_{2x}^2 \frac{32e^{z^2}}{y^2+4}\,dz\,dx\,dy + \int_0^1 \int_{2-2x}^2 \int_{2x}^2 \frac{32e^{z^2}}{y^2+4}\,dz\,dy\,dx$$

10. (10pts) (a) Convert the integral to an equivalent integral in spherical coordinates.

$$\int_{\theta=0}^{\pi/2} \int_{r=0}^{2/\sqrt{\cos^2\theta+2\sin^2\theta}} \int_{z=0}^{r\sin\theta} r\,dz\,dr\,d\theta + \int_{\theta=0}^{\pi/2} \int_{r=2/\sqrt{\cos^2\theta+2\sin^2\theta}}^{2} \int_{z=0}^{\sqrt{4-r^2}} r\,dz\,dr\,d\theta$$

(b) Find the value of the integral in (a).